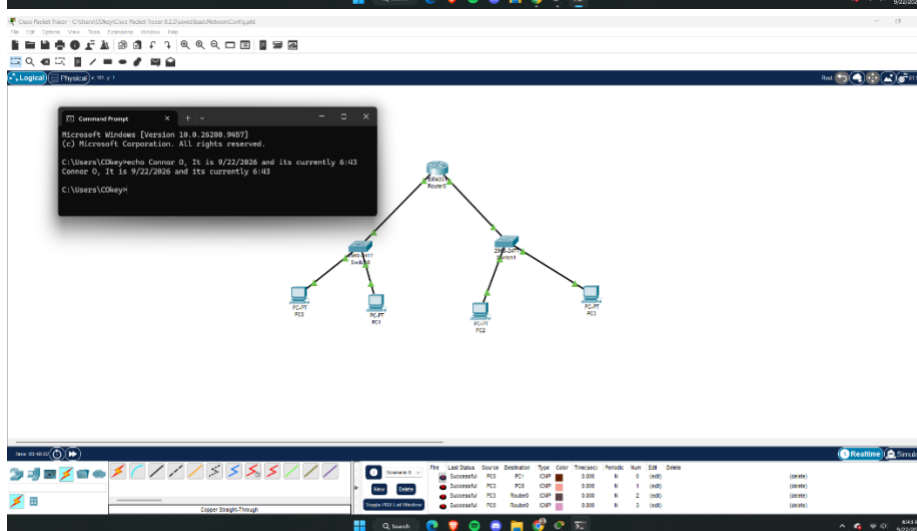
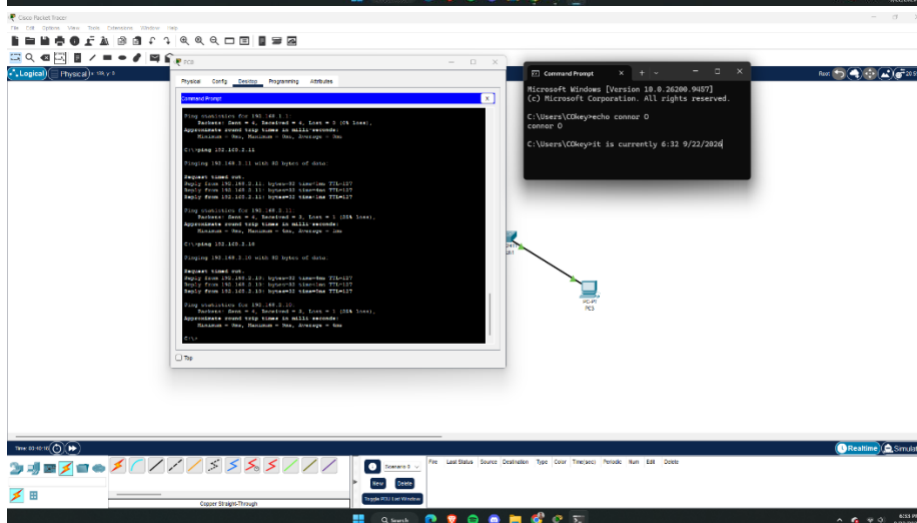
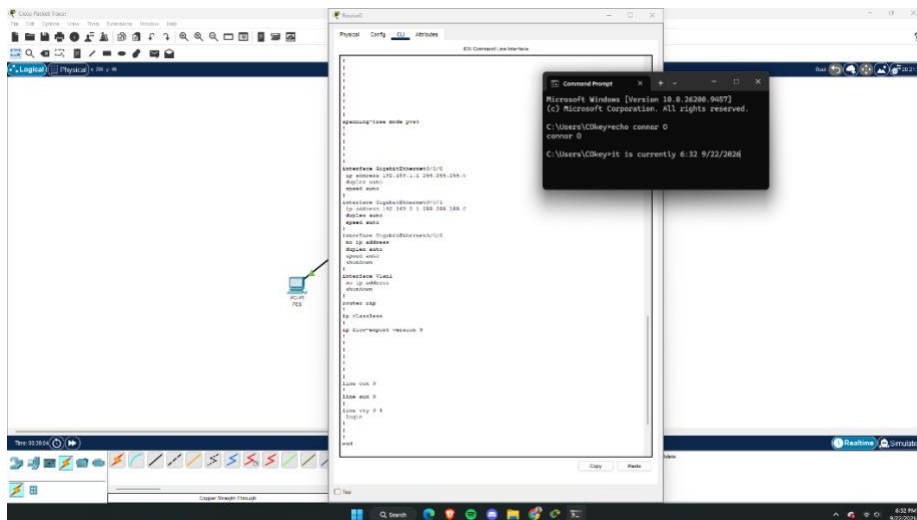




1. What IP addressing scheme did you use?

For the PC's 1-4, their IP's were:

192.168.1.10 255.255.255.0

192.168.1.11 255.255.255.0

192.168.2.10 255.255.255.0

192.168.2.11 255.255.255.0

2. How did you divide your networks (subnets)?

I divided the networks into 192.168.1.1 for the first network which was connected to the GigaEthernet0/0/0, and 192.168.2.1 for the second network which was connected to the GigaEthernet0/0/1.

3. What did you label your devices (PCs, router, etc.)?

I left them as the default names that Cisco gave them. The router was left as Router0, Switch0, and Switch1, and PC0-PC3.

4. What commands did you configure on the router?

This first chunk of code was to configure the first network.

enable

configure terminal interface

GigabitEthernet0/0/0 ip address 192.168.1.1 255.255.255.0

no shutdown

exit

This chunk of code was to configure the second network.

interface GigabitEthernet0/0/1

ip address 192.168.2.1 255.255.255.0

no shutdown

exit

exit

write memory

(Running the commands was confusing for me, so I did use AI to help me configure the Router)

1. What is a **default gateway**?

A default gateway is an IP address configured on the router. When a computer wants data that is outside of the local network, it sends that information to the router. The router then retrieves the data and sends it back to the requesting device.

2. Why was it required for your PCs to communicate with another network?

The default gateway was needed because the devices can't communicate directly with each other since they are on different networks. The router retrieves the data that the original device requested.

1. What problem did you run into?

A problem that I ran into, I swapped the default gateways and the ports the cables were plugged into. So I had no idea what was wrong with it and why it wasn't working.

2. How did you fix it?

I resorted to AI to help me figure it out. It had me run a handful of commands in the command prompts on the computers, switches, and the router. Eventually, the output the command prompt gave me into AI. Eventually, AI figured out that the wires just needed to be switched and then everything would work. So I swapped which ports the wires were plugged into, and the network started working.

3. What part was the most confusing?

Learning how to use Cisco was pretty confusing for me. But configuring the router with the IP addresses was confusing because I didn't know what prompts I needed to input into the router to configure it, which is when I resorted to AI to help me.

I learned a lot from the assignment. It was nice to actually apply what I've been learning from these assignments. And also being able to set up an actual network is nice.